

SEMINAR 2 (Pilot). LIMITS OF SEQUENCES

Let $\{a_n\}$ be a sequence of real numbers. A number a is the **limit** of the sequence $\{a_n\}$, or

$$\lim_{n \rightarrow \infty} a_n = a.$$

if for any $\varepsilon > 0$ there exists a natural number $N \in \mathbb{N}$ depending on ε such that, for every natural number $n > N$, we have $|a_n - a| < \varepsilon$:

$$\forall \varepsilon > 0 \exists N = N(\varepsilon) \quad \forall n > N : |a_n - a| < \varepsilon.$$

In this case the sequence $\{a_n\}$ is said to **converge** to or **tend** to the limit a .

If a sequence converges to some limit, then it is **convergent**; otherwise it is **divergent**.

1. Prove, by definition, that

$$(a) \lim_{n \rightarrow \infty} \frac{3}{n+2} = 0; \quad (b) \text{ (HW) } \lim_{n \rightarrow \infty} \frac{4}{2n-1} = 0; \quad (c) \text{ (HW) } \lim_{n \rightarrow \infty} \frac{(-1)^n}{3\sqrt{n}+11} = 0;$$

2. Prove, by definition, that

$$(a) \lim_{n \rightarrow \infty} \frac{3n+1}{n+2} = 3; \quad (b) \lim_{n \rightarrow \infty} \frac{5n+(-1)^n}{3n+4} = \frac{5}{3}; \quad (c) \text{ (HW) } \lim_{n \rightarrow \infty} \frac{3n-2}{4n+5} = \frac{3}{4}.$$

3. Prove, by definition, that

$$(a) \lim_{n \rightarrow \infty} \frac{\sqrt{n}+4}{2\sqrt{n}+6} = \frac{1}{2}; \quad (b) \text{ (HW) } \lim_{n \rightarrow \infty} \frac{3\sqrt{n}-2}{2\sqrt{n}+4} = 3/2; \quad (c) \text{ (HW) } \lim_{n \rightarrow \infty} \frac{7n^2-4}{4n^2+2} = 7/4;$$

4. Define a divergent sequence and prove, by definition, that the sequence $\{(-1)^n\}$ is divergent.

5. (HW) Prove that the sequence $\{(-1)^n n - 3\}$ is divergent.

6. Prove that if $x_n \geq 0$ then $\lim_{n \rightarrow \infty} x_n = L$ implies $\lim_{n \rightarrow \infty} \sqrt{x_n} = \sqrt{L}$.

7. Prove that (a) $\lim_{n \rightarrow \infty} \frac{n}{a^n} = 0$ if $a > 1$; (b) (HW) $\lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0$, $a > 1$, $k \in \mathbb{N}$.

8. Suppose $\forall n \in \mathbb{N}$ the inequality $x_n < C$ holds and A is a limit point of the sequence $\{x_n\}$. Does inequality $A < C$ hold? Does inequality $A \leq C$ hold? Justify your answer.

9. (HW) Suppose $\lim_{n \rightarrow \infty} x_n = L < b$. Prove that $\exists N \in \mathbb{N} \forall n \geq N : x_n < b$.

10*. (HW) Suppose $\lim_{n \rightarrow \infty} x_n = L$. Prove that $\lim_{n \rightarrow \infty} \frac{x_1 + x_2 + \cdots + x_n}{n} = L$.

Memorize the following limits:

$$\lim_{n \rightarrow \infty} a^n = 0, 0 < a < 1; \quad \lim_{n \rightarrow \infty} \frac{n^k}{a^n} = 0, a > 1, k \in \mathbb{N}; \quad \lim_{n \rightarrow \infty} \sqrt[n]{a} = 1; \quad \lim_{n \rightarrow \infty} \sqrt[n]{n} = 1.$$